Final Exam Study Guide

EC325: Econometrics | Colby College

Prof. Caraher

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Unit 7: Dummy Variables and Non-Linearities

Dummy Variables

- A **dummy variable** translates a qualitative characteristic into 0/1 form: $D = 1$ if the characteristic is present, $D = 0$ otherwise

Interpreting Dummies

- In $score = \beta_0 + \beta_1(hsgpa) + \beta_2(calculus) + \mu$, $\hat{\beta}_2$ is the **difference in intercepts** between the two groups
- Both groups share the same slope; the dummy shifts the regression line up or down

The Dummy Variable Trap

- For a categorical variable with g categories, include only $g - 1$ dummies; the omitted category is the **baseline**
- Including a dummy for every group creates perfect multicollinearity
- Dummy coefficients are always interpreted **relative to the baseline**

Choosing a Baseline

- Changing the baseline does not change the coefficients on other variables
- Pick a baseline that makes interpretation natural (e.g., male as baseline for the gender wage gap)

Quadratic Models

- $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \mu$
- $\beta_2 > 0$: U-shape; $\beta_2 < 0$: inverted-U
- Marginal effect is **not constant**: $\frac{\partial Y}{\partial X} = \beta_1 + 2\beta_2 X$
- Peak/trough at $x^* = -\frac{\beta_1}{2\beta_2}$

Log Transformations

- Difference in logs   proportional change: $\ln(Y_1) - \ln(Y_0) \approx \frac{Y_1 - Y_0}{Y_0}$
- Approximation works well for small coefficients ($|\beta| < 0.20$)
- Exact formula: $100 \times (e^{\beta} - 1)\%$
- Don't log: variables with zeros/negatives, rates/percentages, dummies

Functional Form Reference

Model	Equation	Interpretation of $\hat{\beta}_1$
Level-Level	$Y = \beta_0 + \beta_1 X$	1-unit $\uparrow X \rightarrow \beta_1$-unit $\Delta\ Y$
Log-Level	$\ln(Y) = \beta_0 + \beta_1 X$	1-unit $\uparrow X \rightarrow (100\beta_1)\%\ \Delta\ Y$
Level-Log	$Y = \beta_0 + \beta_1 \ln(X)$	$1\% \uparrow X \rightarrow \beta_1/100$ unit $\Delta\ Y$
Log-Log	$\ln(Y) = \beta_0 + \beta_1 \ln(X)$	$1\% \uparrow X \rightarrow \beta_1\%\ \Delta\ Y$ (elasticity)
Quadratic	$Y = \beta_0 + \beta_1 X + \beta_2 X^2$	Marginal effect $= \beta_1 + 2\beta_2 X$
Dummy	$Y = \beta_0 + \beta_1 D$	Difference between $D = 1$ and baseline
Dummy in Log-Y	$\ln(Y) = \beta_0 + \beta_1 D$	$(100\beta_1)\%$ diff. from baseline

Unit 8: Interaction Effects

Why Interactions?

- Dummies alone shift the **intercept** and force both groups to share the same slope
- Interactions let the effect of one variable **depend on** the value of another
- Built by including the **product** of two variables on the RHS

Two Dummies Interacted

- $\ln(wage) = \beta_0 + \beta_1(female) + \beta_2(married) + \beta_3(female \times married) + \mu$
- The interaction equals 1 only when both dummies equal 1
- The baseline group is where both dummies equal 0 (here, unmarried men)

Group	female	married	Predicted $\ln(wage)$
Unmarried men (baseline)	0	0	β_0
Unmarried women	1	0	$\beta_0 + \beta_1$
Married men	0	1	$\beta_0 + \beta_2$
Married women	1	1	$\beta_0 + \beta_1 + \beta_2 + \beta_3$

- β_3 is the **additional** effect of being both female and married, beyond the sum of the individual effects

Dummy × Continuous

- $\ln(wage) = \beta_0 + \beta_1(female) + \beta_2(educ) + \beta_3(female \times educ) + \mu$
- Men: $\ln(wage) = \beta_0 + \beta_2(educ)$
- Women: $\ln(wage) = (\beta_0 + \beta_1) + (\beta_2 + \beta_3)(educ)$
- β_1 = intercept difference; β_3 = **slope difference**
- The interaction lets lines have different slopes — they can converge, diverge, or cross

Total Effects

- The total effect of a variable includes **all terms where it appears**
- Effect of education: $\beta_2 + \beta_3 \cdot female$ (different for men vs. women)
- Effect of being female: $\beta_1 + \beta_3 \cdot educ$ (depends on education)

Unit 9: Heteroskedasticity

What Heteroskedasticity Is

- GM5 (homoskedasticity): $Var(\mu \mid x_1, \dots, x_k) = \sigma^2$ — constant error variance
- **Heteroskedasticity:** the error variance changes with x
- Common in economic data: income & spending, firm size & profits, experience & wages

What Breaks

- OLS coefficients are still **unbiased** (depends only on GM 1–4)
- But standard errors are wrong — typically too small
- t-statistics too large, p-values too small → **Type I errors inflated**

The Fix: Robust SEs

- Heteroskedasticity-robust SEs weight each squared residual by $(x_i - \bar{x})^2$
- Coefficients unchanged; only SEs differ

When Not Needed

- Robust SEs are slightly larger than OLS SEs under homoskedasticity
- Cost is small; benefit when needed is large
- Default to Heteroskedasticity-robust SEs

Unit 10: Binary Outcomes

The Linear Probability Model (LPM)

- OLS with a dummy/binary y estimates a **probability**: $\hat{y} = \hat{P}(y = 1 \mid \mathbf{x})$
- $\hat{\beta}_j$ = change in the **probability** of $y = 1$ for a one-unit increase in x_j
- **Interpret in percentage points, not percent.** A coefficient of -0.15 is a 15 *percentage point* drop, not 15%

Two Problems with the LPM

1. **Always heteroskedastic:** $Var(\mu \mid \mathbf{x}) = p(\mathbf{x})[1 - p(\mathbf{x})]$, which depends on $\mathbf{x}$. Fix with robust SEs.
2. **Out-of-bounds predictions:** straight line can produce $\hat{p} < 0$ or $\hat{p} > 1$

Logit Regression

- Wraps the linear index in the logistic function $G(z) = e^z/(1 + e^z)$, so predictions stay in $(0, 1)$
- Estimated by **MLE**, not OLS

Three Ways to Interpret Logit Coefficients

1. **Log-odds:** $\hat{\beta}_j$ is the change in log-odds. Mostly useful for *direction* (positive/negative).
2. **Odds ratios:** $\exp(\hat{\beta}_j)$. OR > 1 increases the odds; OR < 1 decreases.
3. **Marginal effects:** the change in **probability** for a one-unit change in x_j. Effect depends on where you start.

Marginal Effect at the Mean

- Standard practice: evaluate the marginal effect at the mean of all explanatory variables
- Compute the predicted probability at the mean, then at the mean with x_j increased by one unit; the difference is the marginal effect

LPM vs. Logit

- The logistic function is nearly linear in the middle of its range
- Near $p \approx 0.5$, LPM and logit give similar marginal effects
- They diverge mostly at the tails (p near 0 or 1)
- LPM useful when interpretation is the goal; Logit is best when prediction is the goal

Unit 11: Panel Data and Fixed Effects

Panel Data

- **Cross-sectional:** units observed at one point in time
- **Panel:** same units observed at multiple points in time
- Panel structure lets us control for time-invariant unobservables

Between vs. Within Variation

- **Between:** comparing different units (confounded by all the ways units differ)
- **Within:** comparing each unit to itself over time (unit-level confounders cancel)
- FE methods exploit within variation

The Fixed Effects Model

$$y_{it} = \beta_0 + \beta_1 x_{it} + a_i + \mu_{it}$$

- a_i is the **unit fixed effect** — captures all time-invariant characteristics of unit i

The Within Transformation

- For each unit, subtract the time-average of every variable: $\ddot{y}_{it} = y_{it} - \bar{y}_i$
- a_i drops out (since $a_i - \bar{a}_i = 0$)
- β_1 is identified from within-unit variation only

Two-Way Fixed Effects

- Add **time fixed effects** to absorb shocks common to all units in a period:

$$y_{it} = \beta_1 x_{it} + a_i + \tau_t + \mu_{it}$$

- Equivalent to including dummies for each unit and each time period (minus the omitted ones)

What FE Do and Don't Control For

- **Unit FE:** anything specific to a unit that is constant over time
- **Time FE:** anything that hits all units equally in a given period
- **Neither:** unit-specific, time-varying confounders (must be controlled for explicitly)
- You can't include time-invariant variables — they're collinear with the unit FE

Unit 12: Difference-in-Differences

Motivation

- DiD uses **control units** rather than (just) **control variables**
- Strategy: find a **natural experiment** where treatment is as good as randomly assigned

The ATT and the Counterfactual

- Average Treatment Effect on the Treated:

$$ATT = E[Y_{i1}(1) \mid D_i = 1] - E[Y_{i1}(0) \mid D_i = 1]$$

- We observe the first term; the second is the unobservable counterfactual

Parallel Trends Assumption

- Without treatment, the treated group would have followed the **same trend** as the control group:
$$E[Y_{i1} - Y_{i0} \mid D_i = 1] = E[Y_{i1} - Y_{i0} \mid D_i = 0]$$
- This is the **core assumption** of DiD — and it's not testable directly

The DiD Estimator

$$ATT = \underbrace{(E[Y_{i1} \mid D_i = 1] - E[Y_{i0} \mid D_i = 1])}_{\text{trend, treated}} - \underbrace{(E[Y_{i1} \mid D_i = 0] - E[Y_{i0} \mid D_i = 0])}_{\text{trend, control}}$$

The DiD Regression

$$y_{it} = \beta_0 + \beta_1 D_i + \beta_2 P_t + \delta(D_i \times P_t) + \mu_{it}$$

- $D_i = 1$ for treated; $P_t = 1$ for post-period
- δ on the interaction term is the DiD estimate (the ATT)

Group	D_i	P_t	Predicted $\hat{y}_{it}$
Control, Pre	0	0	β_0
Control, Post	0	1	$\beta_0 + \beta_2$
Treated, Pre	1	0	$\beta_0 + \beta_1$
Treated, Post	1	1	$\beta_0 + \beta_1 + \beta_2 + \delta$

Two-Way Fixed Effects DiD

$$y_{it} = \delta D_{it} + \alpha_i + \tau_t + \mu_{it}$$

- $D_{it} = 1$ only for treated units in post-treatment periods
- In the 2×2 case, gives the same $\hat{\delta}$ as the standard DiD regression
- Generalizes to many units and many time periods

Choosing Good Controls

- DiD estimate is only as good as the implied counterfactual
- **Levels don't matter; trends do.** A control with a different level but the same pre-trend is fine
- Direction of bias from bad controls:
  - Control declines faster than treated → bias **toward zero**
  - Control declines slower or rises → bias **away from zero**

Two Approaches to Controls

- **Intuition-driven:** based on theory, geography, institutional knowledge (Card & Krueger 1994: NJ vs. PA fast-food on the border)
- **Data-driven:** algorithms that match on observables (synthetic controls, propensity scores, etc.)

Common Threats to DiD

- **Differential pre-trends:** groups already diverging before treatment
- **Anticipation effects:** units change behavior before treatment arrives
- **Spillovers:** treatment affects control units too
- **Composition changes:** the mix of units changes over time

Unit 13: Advanced DiD

IPW-DiD

- When treated and control units differ systematically, parallel trends may fail
- IPW reweights controls to look more like treated units on observables

The IPW-DiD Algorithm

1. Estimate propensity scores from a logit on pre-treatment covariates: $P(D_i = 1 \mid \mathbf{x})$
2. Compute weights: $IPW_i = D_i + (1 - D_i)\frac{\hat{P}_i}{1-\hat{P}_i}$
3. Estimate the TWFE DiD using IPW_i as observation weights

What the Weights Do

- Treated units get weight 1
- Controls that *look like* treated units (high $\hat{P}_i$) get **large** weights
- Controls that look very different get **small** weights
- IPW-DiD is only as good as your propensity score model

Dynamic DiD

- Estimates separate treatment effects for each post-treatment period
- Uses **relative time** k: $k = 0$ in the year of treatment, $k = -1$ one year before, $k = 1$ one year after

$$y_{it} = \sum_{j=-J}^{-2} \beta_j D_{it}^j + \sum_{k=0}^{K} \beta_k D_{it}^k + \alpha_i + \tau_t + \mu_{it}$$

- $D_{it}^k = 1$ if the unit is treated and relative time equals k
- **Omit** the dummy for $k = -1$ as the reference period

Interpreting Dynamic DiD

- Each $\hat{\beta}_k$ is the treatment effect at relative time k, relative to the period just before treatment
- Pre-treatment coefficients ($k < -1$) test parallel trends — they should be 0
- Post-treatment coefficients trace out the dynamic effect over time
- Present results as an **event study plot** (coefficient + 95% CI vs. relative time)

Cluster-Robust Standard Errors (CRSEs)

- A **cluster** is a group of observations whose errors are correlated with each other but independent across groups (students within classrooms, individuals within states)
- Within-cluster correlation violates OLS independence
- Ignoring it makes SEs too small → massive Type I error inflation
- HC1 corrects for heteroskedasticity but **not** for within-cluster correlation

Choosing the Cluster Level

- Coefficients are unchanged; only SEs change (typically larger)
- **Rule:** cluster at the level of treatment assignment
- Need at least 20–50 clusters; when in doubt, cluster at a broader level

Unit 14: Instrumental Variables

The Endogeneity Problem

- OLS requires $E[\mu \mid X] = 0$
- When $Cov(X, \mu) \neq 0$, X is **endogenous** and OLS is biased
- Three classic sources:
  1. Omitted variable bias (e.g., ability bias)
  2. Selection bias (e.g., healthier people buy insurance)
  3. Reverse causality (e.g., crime $\rightarrow$ police hiring)

The IV Idea

- Variation in X has two parts:
  - Endogenous (correlated with μ) — biases OLS
  - Exogenous (unrelated to μ) — what we want
- An **instrument** Z isolates the exogenous variation in X

The Two IV Assumptions

1. **Relevance:** $Cov(Z, X) \neq 0$ — the instrument moves X. Testable via the first-stage F.
2. **Exclusion restriction:** $Cov(Z, \mu) = 0$ — Z affects Y only through X. Not testable; defended with economic reasoning.

Angrist & Evans (1998)

- Question: causal effect of additional children on mothers' labor force participation
- Endogeneity: career orientation drives both fertility and work
- Instrument: **sex composition of the first two children** (parents prefer mixed-sex; same-sex parents are more likely to have a third)
- Relevance: same-sex parents are 6 pp more likely to have a third child
- Exclusion: child sex is biological chance

Two-Stage Least Squares (2SLS)

- **Stage 1:** $X_i = \pi_0 + \pi_1 Z_i + \nu_i$, get $\hat{X}_i$
- **Stage 2:** $Y_i = \beta_0 + \beta_1 \hat{X}_i + \varepsilon_i$
- $\hat{\beta}_1^{2SLS}$ is the IV estimate
- Running the two stages by hand gives correct *coefficients* but wrong *standard errors* — use a proper 2SLS estimator that adjusts SEs for first-stage uncertainty

Multiple Instruments

- More instruments → more first-stage variation, smaller SEs
- More instruments than endogenous variables → **overidentified** model
- Each instrument needs its own exclusion restriction defended
- A few strong instruments usually beat many weak ones

Weak Instruments

- A weak instrument barely moves $X \rightarrow$ wild swings in $\hat{\beta}^{2SLS}$, unreliable inference
- Standard diagnostic: **first-stage F-statistic**
- Old rule of thumb: $F > 10$ (Stock & Yogo 2005)
- Recent evidence (Lee, McCrary, Moreira & Porter 2022): need F closer to 100 for reliable inference

Diagnostics

- **First-stage F (homoskedastic):** standard relevance check
- **Robust first-stage F (Wald form):** preferred when clustering or with heteroskedasticity
- **Wu-Hausman test:** compares OLS and 2SLS coefficients to check whether X is in fact endogenous (a sanity check that IV is doing necessary work)

Unit 15: Regression Discontinuity

The RD Idea

- Modern life is full of rigid cutoffs (turn 21, score 70+, vote share > 50%)
- At the cutoff, treatment flips on/off abruptly
- Units just above and just below the cutoff are nearly identical except for treatment
- Rigid rules generate natural experiments

Vocabulary

- **Running variable:** the continuous variable that determines treatment (age, test score)
- **Cutoff (threshold):** the value at which treatment flips
- **Sharp RD:** treatment is a deterministic function of the running variable
- **Fuzzy RD:** the cutoff shifts the *probability* of treatment (handled with IV — beyond this course)

MLDA Example

- Carpenter & Dobkin (2009): does the minimum legal drinking age save lives?
- Compare death rates just below and just above age 21
- The "jump" at the cutoff is the causal effect

The Sharp RD Regression

$$Y_a = \alpha + \rho D_a + \gamma a + e_a$$

- $D_a = 1$ if $a \geq a_0$
- γa controls for the smooth trend in the running variable
- ρ — the **vertical jump** at the cutoff — is the causal estimate

Handling Nonlinearity

- A naive linear fit can mistake a smooth curve for a jump
- Two approaches:
  - **Parametric:** add polynomials and interactions; let slope/curvature differ on each side
  - **Nonparametric:** restrict to a narrow window where any smooth curve looks linear
- Best practice: report several specifications and show estimates are stable
- Don't use polynomials higher than quadratic — they invent jumps near the boundaries

Bandwidth: a Bias-Variance Tradeoff

- Narrow bandwidth: less data → bigger SE, but less room for nonlinearity (low bias, high variance)
- Wide bandwidth: more data → smaller SE, but more room to mistake curvature for a jump (high bias, low variance)